Siamese Correlation Networks for Identifying Gravitationally Lensed Fast Radio Bursts: Real-Data Validation and the Limits Set by Survey Geometry

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***Abstract*—A fast radio burst (FRB) lensed by a foreground galaxy would reach Earth twice, days to months apart, with the two copies differing only by propagation effects. Recognising such copies among the thousands of bursts from repeating sources is a pairwise verification problem with no real positive examples. We build a semi-synthetic benchmark from all 981 public dynamic spectra of the 83 repeating sources in the CHIME/FRB Catalog 2, with real noise, measured signal-to-noise ratios and *impostor* pairs of different bursts that share a propagation imprint. A 259k-parameter Siamese encoder that aligns two feature maps with a differentiable correlation layer before fusing them recovers 0.47 [0.25, 0.66] of lensed pairs at 1% false-positive rate in source-grouped cross-validation, against 0.25 [0.20, 0.30] for the best physics- or feature-based baseline. Ablations over three seeds show that both the alignment layer and impostor training are needed. Because no real lensed pair exists, we validate on real data in three ways: real cross-source pairs serve as true negatives, scored against per-pair simulated twins; two copies of a real burst stand in for lensed pairs; and the whole pipeline is reproduced on a second machine. Finally, we show that survey geometry, not the classifier, limits discovery: a transit telescope records the copies of a repeater's bursts only for the 1.8% of lenses whose delay is close to a whole number of sidereal days, so an all-sky monitor would find about $\sim$50$\times$ more lensed repeaters than CHIME from the same sources.**

***Index Terms*—Siamese networks, pairwise verification, simulation-to-real transfer, signal processing, radio astronomy, gravitational lensing**

I. Introduction

Fast radio bursts are millisecond radio flashes from distant galaxies; some sources repeat thousands of times. If a galaxy lies close to the line of sight, gravitational lensing produces two images of the source, and every burst then arrives twice, separated by a fixed delay of days to months [1], [2]. A confirmed lensed FRB would measure the Hubble constant and test dark-matter models, but none is known [3]–[6].

Finding one is a pairwise verification problem: given two bursts from the same sky position, are they two copies of one burst, or two different bursts from the same repeating source? This is hard for three reasons. There are no real positive examples, so any classifier is trained on simulation and must be validated on real data without labels. The negatives are adversarial: different bursts of one source can share a spectral envelope imposed by plasma lensing, or a scintillation pattern, and look like copies. And the decision must hold at very low false-positive rates, because the prior probability of lensing is about 10^{-3} per source.

Lensed-pair classifiers exist for gravitational waves [7], [8], where Siamese networks fuse paired features by their absolute difference and product. FRB lensing searches have so far used auto- and cross-correlation statistics [3], [4]. Our contributions are:

1) a semi-synthetic pair benchmark built from all public CHIME/FRB Catalog 2 repeater spectra, with impostor negatives and frozen, hash-checked evaluation sets (Section III);
2) a Siamese correlation network that aligns feature maps before fusing them, with an ablation study (Sections IV, VI);
3) a validation protocol for a detector without real positives: true-negative transfer with per-pair twins, a real-burst positive proxy, and an independent reproduction (Section V);
4) an analysis of where such a detector can work, which shows that transit telescopes rarely record both copies and that sky coverage, not classifier accuracy, limits discovery (Section VII).

II. Related Work

Siamese verification. Siamese networks decide whether two inputs share an identity by comparing embeddings from one shared encoder, from signature verification [9] to one-shot recognition [10]. Most compare global vectors, which discards the relative position of features; our classic Siamese baseline (5) is of this kind. Correlation layers that compare feature maps over a range of displacements were introduced for optical flow [11]; we use one to estimate and remove the residual time shift between two bursts before comparison.

Lensed-pair identification. For gravitational waves, lensed event pairs have been identified with a DenseNet and gradient boosting [7] and with a Siamese residual network that fuses features by absolute difference and product [8]; the latter notes the need for adaptation to real detector data. For FRBs, lensing searches have used auto- and cross-correlation statistics [3], [4], [6], and machine learning has been used to classify repeating from non-repeating sources [12], not to identify lensed copies.

Simulation-to-real transfer. Detectors trained on simulation are commonly made robust by randomising nuisances during training [13]. Without real positives, transfer must also

be checked on real data; we do so with real negatives, per-pair twins and a real-signal positive proxy.

III. Data and Benchmark

Real inputs. We use the Stokes-I dynamic spectra of all 981 bursts of the 83 repeating sources in CHIME/FRB Catalog 2 [14], at 0.983 ms and 400–800 MHz. Each burst is binned to 256 frequency channels $\times$ 160 time samples around the emission and normalised by its off-pulse noise. The catalogue's fitted burst models serve as noise-free templates (950 eligible), and off-pulse columns of each file, mirror-tiled, serve as real backgrounds (856).

Pair classes. A *lensed* pair places one template twice; a *repeat* pair places two templates of one source; an *unrelated* pair takes templates of two sources; an *impostor* pair is a repeat pair that shares a propagation imprint: member B inherits member A's 25-MHz-smoothed spectral envelope (I1, mimicking a plasma lens), both share one scintillation pattern (I2), or both. Each member is injected at a measured matched-filter S/N on a different real background, with its own time shift (± 10 samples), residual dispersion, scattering and scintillation, so that only burst identity separates lensed from repeat pairs. Single-member statistics cannot tell the classes apart (probe AUC ≤ 0.53).

Splits and frozen sets. Sources are split 60/20/20 into train, validation and test. Validation and test sets of 4000 pairs, and "bright" sets with pair minimum S/N ≥ 25 (the pre-registered headline bin), are regenerated from fixed seeds and checked against stored SHA-256 hashes. The test split stays sealed under a pre-registered blind protocol; all numbers below are development results on sources never used for training.

Metric. Recall of lensed pairs at 1% false-positive rate against repeat and impostor pairs, with 95% intervals from a source-level bootstrap that recomputes the threshold in every resample.

IV. Method

Benchmark ladder. All methods score the same pairs, averaged over both member orders: (1) normalised cross-correlation in the on-pulse box, maximised over ± 24 lags, with 1, 4 or 16 channel blocks; (2) a physics statistic that fits one member to the other over scattering time, residual dispersion, lag and amplitude; (3) gradient-boosted trees (GBM) on seven measured burst properties per member (best of 1–3, the comparator); (4) a two-channel CNN; (5) a classic Siamese network comparing vector embeddings; (6) our Siamese correlation network (Fig. 2); and (7) cross-attention between time–frequency patches.

Model 6. Inputs pass through asinh to compress the dynamic range. The encoder g has four 5×3 convolution blocks with group normalisation, reducing 256×160 to 64 channels $\times\, 16 \times 80$ while keeping two-sample time resolution, because lensed copies differ by residual time shifts. The correlation layer computes the normalised correlation $\rho(\ell)$ of the two maps over lags $|\ell| \leq 12$ encoder steps (± 24 samples), forms softmax weights $w_\ell \propto \exp(\beta \rho(\ell))$ with learned β, and aligns g_B as $\sum_\ell w_\ell\, g_B(\ell)$. Fusion concatenates $|g_A - g_B|$ and $g_A \odot g_B$, two convolution blocks and mean/max pooling follow, and an MLP receives the pooled features with ρ. The model has 259k parameters.

Training. All learned models are trained for 20 000 steps of 64 pairs generated on the fly from training sources (half with S/N floor 10, half 25), with four-class cross-entropy, AdamW (learning rate 10^{-3}, weight decay 10^{-4}), 1000 warm-up steps and cosine decay; the checkpoint is selected on the validation bright set. The lensed score is the log-odds of the lensed class against the other three.

V. Validation Without Real Positives

Grouped folds. Because the validation interval was wide, a pre-registered rule moved the primary comparison to five source-grouped folds: each source is scored by a model that never saw it.

True-negative transfer. Two bursts from different sources cannot be lensed copies, so real cross-source pairs are labelled negatives that can be scored without breaking the blind protocol. For each such pair we simulate five *twins* from the same two templates at their measured S/N, and set the threshold at the twins' 99th percentile; the pre-registered criterion is a real false-positive rate of at most 2%. Twin matching matters: with a threshold from generic simulated pairs, cross-correlation appears to fail below S/N 25 (real rate 6.2, 3.8, 2.4% for its three variants), yet its twins fail just as often (5.6, 4.0, 2.5%) because the real pairs there are brighter than generic ones (median S/N 16.9 vs 13.5). The "failure" is sample composition, not a simulation gap.

Real-burst positives. As a stand-in for lensed pairs, two copies of a real burst's cutout, with its fine structure, are injected with independent nuisances on different backgrounds, alongside template copies and noise-only copies made with the same random draws.

Reproduction. A second machine re-downloaded and re-processed all 981 spectra, rebuilt the injection cache and re-generated the frozen sets, then retrained model 6 and repeated the checks.

VI. Results

Benchmark. Table I shows that every learned model beats the physics and feature baselines, and model 6 leads all of them, by +0.39 recall over the GBM on the fixed split and +0.22 [+0.04, +0.38] in grouped folds. At the 1% threshold its false positives are almost all I1 impostors, which share a spectral envelope by construction; plain repeats are rarely confused. Model 6 classifies 95% of lensed pairs of unseen sources correctly, while its confusion of repeat and unrelated pairs on unseen sources (32% correct vs 86% on seen) shows that part of its capacity memorises source identity; training variants that remove this incentive are in progress.

The gain holds at every S/N (Fig. 3) and is largest in relative terms at the faint end, where model 6 recovers about 0.41 of lensed pairs against 0.12 for the GBM.

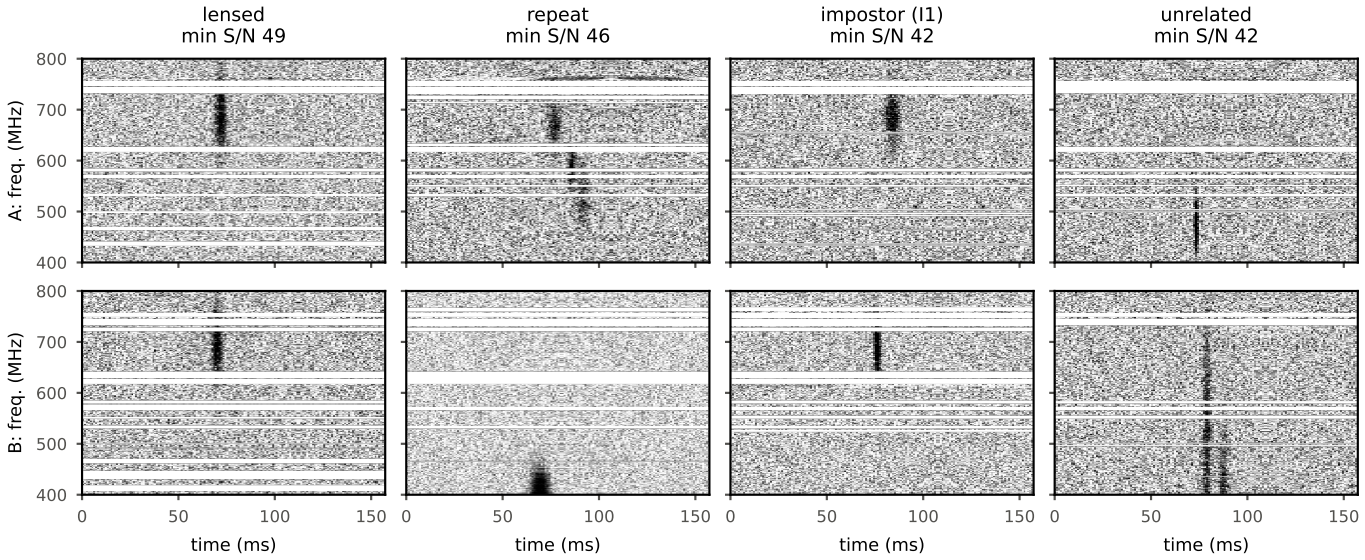


Fig. 1. One validation pair of each class at similar S/N (rows: members A and B; white rows: masked radio-frequency interference). A lensed pair repeats one burst on different real backgrounds with independent nuisances; a repeat pair shows two bursts of one source; an impostor pair shows two bursts of one source where B inherits A's spectral envelope, mimicking a plasma lens; an unrelated pair comes from two sources. The detector must separate the first column from the other three at a 1% false-positive rate.

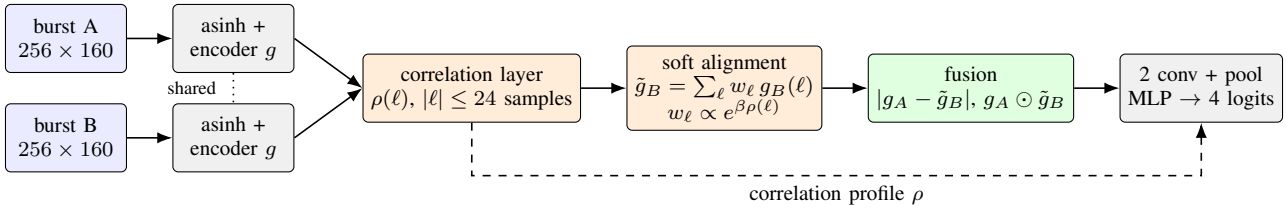


Fig. 2. Model 6, the Siamese correlation network. A weight-shared encoder maps each burst to a $64 \times 16 \times 80$ feature map at two-sample time resolution; a correlation layer over ± 24 samples measures the residual delay and softly aligns B to A with a learned temperature β; the aligned maps are fused by absolute difference and product. The correlation profile also feeds the head. Both directions are computed and averaged, so the score is symmetric in A and B.

TABLE I
RECALL AT 1% FPR, LENSED VS REPEAT + IMPOSTOR, S/N ≥ 25, VALIDATION SOURCES (95% SOURCE BOOTSTRAP).

Method	Fixed split (3 seeds)	Grouped folds (OOF)
(1) Cross-correlation, 16 blocks	0.171 [0.115, 0.245]	–
(2) Physics fit, 16 blocks	0.164 [0.119, 0.235]	–
(3) Feature GBM	0.285 [0.180, 0.401]	0.246 [0.199, 0.297]
(4) Two-channel CNN	0.423 / 0.409 / 0.407	–
(5) Classic Siamese	0.500 / 0.559 / 0.541	–
(7) Cross-attention	0.471 / 0.013[a] / 0.579	–
(6) Ours	**0.701 / 0.698 / 0.684**	**0.467 [0.253, 0.662]**
(6) − (3), paired	+0.394 [+0.226, +0.542]	+0.220 [+0.035, +0.381]
(6) reproduction[b]	0.668 / 0.649 / 0.616	–

[a]Collapsed to a plateau (one of three seeds). [b]Second machine, different data-loader worker count; seed 0 inside the pre-registered range 0.66–0.72, the mean 0.05 below the original seeds.

TABLE II
ABLATIONS OF MODEL 6 (THREE SEEDS EACH): SEED-MEAN RECALL AND PAIRED DIFFERENCE FROM THE FULL MODEL, VALIDATION BRIGHT SET.

Variant	Recall	Difference [95%]
Full model	0.644	–
No alignment (correlation off)	0.557	−0.087 [−0.186, −0.025]
No impostors in training	0.474	−0.170 [−0.254, −0.052]
Hard-negative repeats	0.698	+0.053 [−0.011, +0.082]

Ablations. Removing the alignment, or training without impostors, lowers recall in every seed (Table II). Without impostors the network mistakes envelope-sharing bursts of one source for copies. Oversampling template-similar repeats helps slightly in every seed, but the interval includes zero, and on a closed-form benchmark this scheme biased a learned likelihood ratio low by about 1 dex, so we do not adopt it without a correction.

Transfer to real negatives. On the 3851 real cross-source pairs of the validation sources, all 15 trained models and baselines meet the twin-threshold criterion in both S/N bins once short real windows are filled with off-pulse noise like the simulated backgrounds: model 6 reaches 0.8–1.4% in the headline bin (GBM 0.009), and its real-to-twin rank correlation reaches the twin-to-twin ceiling. Before the fill, model 6 flagged pairs whose windows shared the same blank edges, a format mismatch that a twin test exposed and the fill removed.

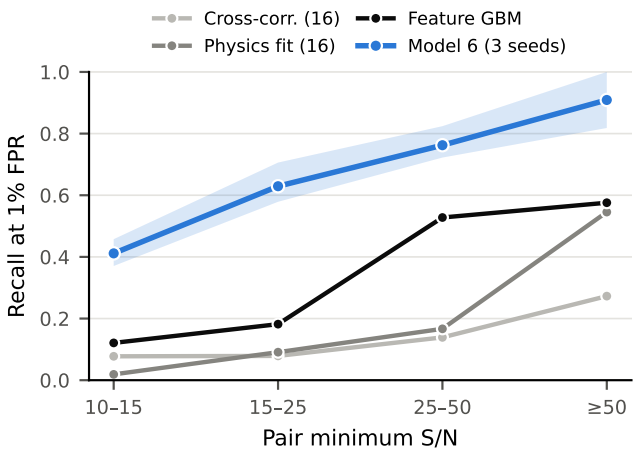


Fig. 3. Recall at 1% FPR by pair minimum S/N on the frozen validation set (529, 330, 108 and 33 lensed pairs per bin). Model 6: mean and range of three seeds trained on the second machine, with checkpoints selected on the separate bright set. Its largest relative gain is at low S/N, the bin with the most pairs.

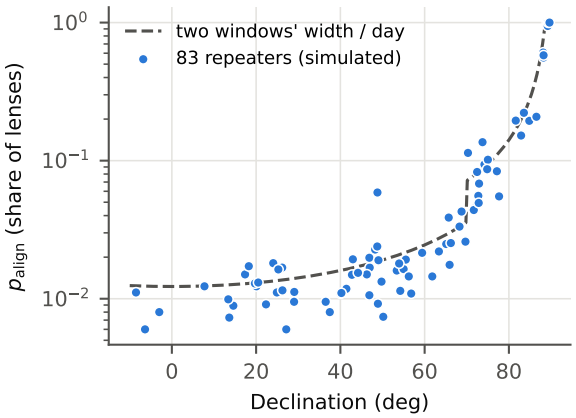


Fig. 4. Share of galaxy-lens delays for which CHIME records the copies of a repeater's bursts, for the 83 Catalog 2 repeaters (points, simulated with each source's own burst times, exposure and redshift) and the analytic window width (dashed).

Repeating this check on the second machine with its own model 6 seeds gives 1.5, 1.3, 1.0% (3851 pairs), all within the limit, with rank correlations 0.91–0.92 against ceilings 0.87–0.89.

Real-burst positives. At injected S/N 25, model 6 recovers 0.663 of real-burst copies against 0.691 of template copies (ratio 0.96 [0.89, 1.06]) and none of the noise-only copies, while the GBM recovers 0.178. The classic Siamese network drops to a ratio of 0.72–0.84, so it relies on template smoothness and model 6 does not. On the second machine, its own model 6 seeds give a real/template ratio of 0.97 [0.91, 1.08].

VII. Where Can Such a Detector Work?

A lensed repeater can also be found without a classifier: a galaxy lens fixes the delay to far better than the 4 ms catalogue timing, so two burst pairs that share one delay are already significant. We calibrated this timing test on burst lists scrambled by whole sidereal days and confirmed its false-alarm rate on independent scrambles (1 of 166,000 held-out scrambles).

Both the classifier and the timing test need the copies to be recorded. CHIME is a transit telescope: it sees a source for minutes per day, and every recorded burst arrives within minutes of transit. The copy of each burst then arrives at the same hour-angle offset, $24\,\mathrm{h} \times \mathrm{frac}(\Delta t / T_{\rm sid})$, so copies are recorded for all bursts or none, and only if the delay is within minutes of a whole number of sidereal days. Simulating singular-isothermal-sphere lenses with each source's redshift, burst times and exposure, this happens for 1.8% of lenses at the median, following the analytic window width (Fig. 4). Repeaters therefore do not escape the transit problem known for one-off bursts [15]. Table III gives the expected number of detectable lensed repeaters: the Catalog 2 sample is expected to hold 5.3×10^{-6}–1.2×10^{-4} (without and with magnification bias), so a null search is the expected outcome; a 10-year transit survey gains only 1.4–1.8×, while an all-sky monitor watching the same sources gains ∼50×. The result is robust to the assumed host dispersion (-19% in yield). For transit surveys the classifier's role is to vet the rare single recorded copy and to discard impostors; its full value lies with wide-field arrays such as BURSTT [16].

TABLE III
Expected lensed repeaters among the 83 Catalog 2 sources (γ: magnification-bias slope).

Survey	$\sum\tau$	$N_{\rm exp}$ (timing)	$N_{\rm exp}$ (≥ 1 copy)
CHIME, Catalog 2	2.9×10^{-3}	5.3×10^{-6}	3.2×10^{-5}
All-sky, same sources	2.9×10^{-3}	2.6×10^{-4}	6.3×10^{-4}
CHIME, Catalog 2, $\gamma=1.5$	1.8×10^{-2}	1.2×10^{-4}	4.9×10^{-4}
All-sky, same sources, $\gamma=1.5$	1.8×10^{-2}	6.1×10^{-3}	9.3×10^{-3}
CHIME-like, 10 yr	2.8×10^{-3}	7.4×10^{-6}	4.9×10^{-5}
All-sky, 10 yr	2.8×10^{-3}	3.7×10^{-4}	6.8×10^{-4}
CHIME-like, 10 yr, $\gamma=1.5$	1.8×10^{-2}	2.2×10^{-4}	6.3×10^{-4}
All-sky, 10 yr, $\gamma=1.5$	1.8×10^{-2}	7.4×10^{-3}	9.6×10^{-3}

VIII. Discussion and Limitations

Simulated false-positive rates rank methods, but deployment thresholds must come from real negatives; twin matching is what makes such a check interpretable, because it separates simulation gaps from differences in sample composition. Our results are development results: the test split is sealed until the team signs off the propagation model, and the reproduction found model 6 about 0.05 recall below the original seeds on a second machine, for reasons we have not isolated. The lens model is a singular isothermal sphere, impostor strength rests on a simulation choice, and the validation set has only 26 sources.

IX. Conclusion

A small Siamese network that aligns before it compares identifies lensed copies of real FRB spectra far better than physics and feature baselines, passes real-data checks designed

for a detector without real positives, and reproduces on a second machine. The same analysis shows that transit geometry, not classification accuracy, is the bottleneck for finding lensed repeaters, which points such detectors to wide-field radio arrays.

Reproducibility. All inputs are public (CHIME/FRB Catalog 2, DOI 10.11570/25.0066). Code, frozen-set manifests and hashes, pre-registered plans and every result update are in the project repository (link withheld for review).

[Test-split result after team sign-off.]

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